THE (SO-CALLED) BREEDERS EQUATION

This will be shown for discrete generations. Start with mass (phenotypic) selection: $\Delta G = ih^2\sigma_P / L$ where ΔG is the gain per year, *i* is the intensity of selection, i.e. standardised selection differential, h^2 is the heritability and σ_P is the phenotypic standard deviation, and *L* is the generation interval in years. One of the most basic assumptions for most of quantitative genetic theory is that the regression of breeding value (g_i) on phenotype (y_i) is $g_i = \bar{g} + h^2(y_i - \bar{y}) + e$ (see QG course or Falconer & Mackay). If selection is made on y_i with intensity *i* the mean of the selected individuals is $\bar{y}^* = \bar{y} + i\sigma_P$. Therefore assuming the linear regression is a good predictor for all subsets (and we do assume it) the regression predicts $\bar{g}^* = \bar{g} + h^2 i\sigma_P$ i.e. the change in genetic merit is $h^2 i\sigma_P$. If it takes *L* years to carry out selection and produce the next generation then $\Delta G = ih^2\sigma_P / L$ as ΔG is the gain per <u>year</u>.

Now generalise to any index of merit, i.e. $\hat{g}_i$, being used as a predictor of genetic merit for selection, such as an EBV. Let the variance of the index be $\mathrm{var}(\hat{g}_i) = \sigma_I^2$ and the variance of the true breeding values be $\mathrm{var}(g_i) = \sigma_A^2$. Let the correlation between the index and the TBV be the <u>accuracy</u>, *r*, so $r = cov(\hat{g}_i, g) / (\sigma_A \sigma_I)$. Construct a regression of g_i on $\hat{g}_i$ so $g_i = \bar{g} + b(\hat{g}_i - \bar{\hat{g}}_i) + e$, this is the same idea as in mass selection where the predictor was just the phenotype. The slope $b = cov(\hat{g}_i, g) / var(\hat{g}_i) = r\sigma_A\sigma_I / \sigma_I^2 = r\sigma_A / \sigma_I$. Now after selecting on the index with s.d. σ_I with intensity *i* (instead of phenotype with s.d. σ_P) $\bar{g}^* = \bar{g} + bi\sigma_I$ and the change in genetic merit is $bi\sigma_I = (r\sigma_A / \sigma_I)\, i\sigma_I = ir\sigma_A$. From which the results follows

$$\Delta G = ir\sigma_A / L$$

i.e. the 'rate of gain per year' = ('intensity' x 'accuracy' x 'genetic s.d.') / 'generation interval'

This is a general formula for any predictor of merit assuming the linear regression of g_i on $\hat{g}_i$ is a good fit, including the tails of the distribution. The formula for mass selection can be obtained by noting that *r* = *h* (see below), and $\sigma_A = h\sigma_P$.

When the individual has only its own phenotype for the trait of interest (P), and no other recorded phenotypes or recorded relatives the best estimate of the BV is $EBV = h^2 P$. The accuracy is the correlation between EBV and TBV, i.e. $cor(\hat{g}, g) = cov(\hat{g}, g) / \sqrt{var(\hat{g}) var(g)}$. These terms are:

$cov(\hat{g}, g) = cov(h^2 P, g) = h^2 cov(P, g) = h^4 var(P)$; $var(\hat{g}) = var(h^2 P) = h^4 var(P)$; and $var(g) = h^2 var(P)$.

These results come from the basic QG model that $P = g + e$ where g and e are independent, and the definition $h^2 = var(g) / var(P)$. Putting all these together, gives the accuracy as $h^4 var(P) / (h^3 var(P) = h$. This is a result worth remembering, the accuracy of a simple phenotype is the square root of the heritability.